

Numerical Verification Supplement

1 Numerical Diagnostics and Galerkin Verification

This supplement records finite-dimensional numerical diagnostics for the calibrated shorting mechanism. It includes only the verification material tied to the paper’s Galerkin consistency result, the exact calibration result, the finite-part regime separation, and the admissibility boundary. The proof appendix does not depend on these computations. The reproducible source code and execution scripts are available at <https://anonymous.4open.science/r/gaussian-hilbert-6CF1>.

Verification Map

- Section 1.1: Galerkin truncation and variational shorting consistency.
- Section 1.2: canonical variance scaling.
- Section 1.3: finite-part regime separation in a three-node Gaussian chain.
- Section 1.4: admissibility boundary stress test.
- Section 1.5: sparse OU/GFF precision-operator implementation and comparison.
- Section 1.6: BigFloat/Float64 agreement audit for residual magnitudes.

1.1 Galerkin Verification and Ill-Conditioning

We illustrate the Galerkin convergence asserted by the paper’s Galerkin consistency result against a finite reference on an admissible model. The Hilbert space is $\ell^2(\mathbb{N})$ with coordinate basis $\{\phi_j\}$. The drift is $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where $\mathcal{A}_0\phi_j = -(1 + 0.01j^2)\phi_j$ and $\mathcal{K}\phi_1 = \sum_{j \geq 2} 12j^{-3}\phi_j$ is a bounded

rank-one nilpotent perturbation. The noise is $\mathcal{D}\phi_j = j^{-3/2}\phi_j$, which is trace-class. The regularized conditioning acts at $k = 1$ with $\varepsilon_0 = 1$. Truncated covariances are compared against a finite reference $N_{\text{ref}} = 1500$.

Table 1: Truncation convergence diagnostics.

Diagnostic	Value	Expected
$\ \mathcal{D} - \Pi_n \mathcal{D} \Pi_n\ _{\mathcal{S}_1}$ slope	-0.499	$\rightarrow 0$
$\ \Sigma_n - \Sigma_{\text{ref}}\ _{\mathcal{S}_1}$ slope	-2.683	$\rightarrow 0$
$\ \Sigma_{n,\lambda} - \Sigma_{\text{ref},\lambda}\ _{\mathcal{S}_1}$ slope	-2.683	$\rightarrow 0$
$ S_{1,n,\lambda} - S_{1,\text{ref},\lambda} $ slope	-5.504	$\rightarrow 0$
$m_2^{\text{Schur},(1000)}$ extrapolated	0.5000000	0.5
$ m_2^{\text{Schur},(1000)} - \varepsilon_0/2 $	6.87×10^{-14}	0

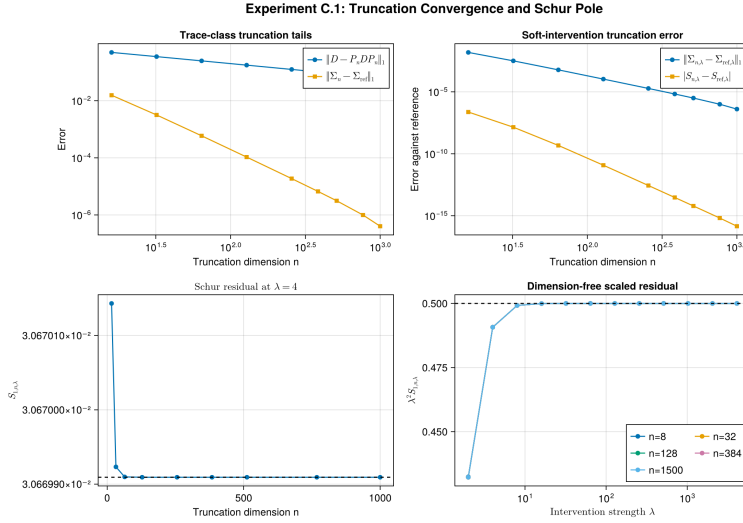


Figure 1: Trace-norm and residual errors decay in n ; the extrapolated $m_2^{\text{Schur},(1000)}$ differs from $\varepsilon_0/2$ by 6.87×10^{-14} .

The variational operator sequence converges, but computing the truncated Schur complement by naive inversion,

$$S^{(n)} = \Sigma_{EE} - \Sigma_{EI}^{(n)} (\Sigma_{II}^{(n)})^{-1} \Sigma_{IE}^{(n)},$$

fails in generic Galerkin bases. The internal block is a trace-class covariance, so its truncated matrix representation becomes severely ill-conditioned as n

grows. The diagnostics show that for $n \geq 400$ the condition number $\kappa(\Sigma_{II}^{(n)})$ grows as n^6 and formally exceeds the 10^{16} IEEE 754 Float64 precision limit.

1.2 Canonical Scaling Diagnostic

We validate the canonical normalization in the paper’s exact calibration result through its decoupled reference channel. With incoming target coordinate blocks and off-diagonal drift coupling set to zero, the target covariance is exactly

$$\Sigma_{\lambda,22} = d_{\lambda}/(2\lambda).$$

Four candidate target-diffusion families are tested:

1. **Canonical:** $d_{\lambda} = \varepsilon_0/\lambda$.
2. **Slow:** $d_{\lambda} = \varepsilon_0$.
3. **Fast:** $d_{\lambda} = \varepsilon_0/\lambda^2$.
4. **Perturbed:** $d_{\lambda} = (\varepsilon_0/\lambda)(1 + \lambda^{-2})$.

Table 2: Exact-calibration ablation. Only the canonical family satisfies the exact finite- λ covariance condition.

Family	$\lambda d_{\lambda}/\varepsilon_0$	defect $ 2\lambda^2\Sigma_{\lambda,kk}/\varepsilon_0 - 1 $
Canonical	1	0
Slow	λ	$\lambda - 1 \rightarrow \infty$
Fast	λ^{-1}	$ 1 - \lambda^{-1} \rightarrow 1$
Perturbed	$1 + \lambda^{-2}$	λ^{-2}

1.3 Three-Node Gaussian Chain

We evaluate the Gaussian Fredholm finite-part functional $\mathfrak{J}_{\lambda}$ on a three-node linear Ornstein–Uhlenbeck SDE,

$$X_1 \rightarrow X_2 \rightarrow X_3,$$

where X_2 is the target coordinate subject to regularized conditioning of strength $\lambda \in \{2^1, \dots, 2^{12}\}$, with $x_2^* = 0$ and $\varepsilon_0 = 1$. The drift and diffusion are

$$A_{\lambda} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & a_{32} & -1 \end{pmatrix}, \quad D_{\lambda} = \text{diag}(1, \varepsilon_0/\lambda, 1).$$

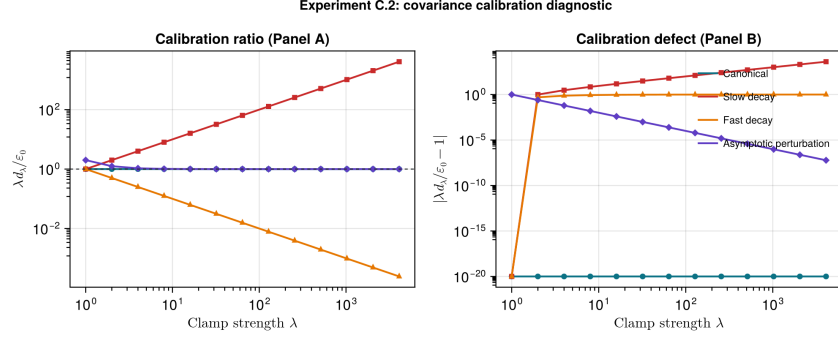


Figure 2: The perturbed family has the correct limiting ratio but fails the exact finite- λ covariance condition. Exact zero defects are displayed at 10^{-20} .

The target mean forcing b_2 controls the target mean shift $m_{\lambda,2} = b_2/\lambda$.

1. **Case A, decoupled-centered:** $a_{32} = 0$, $b_2 = 0$, and $\mathfrak{J}_\lambda = 0$ exactly.
2. **Case B, coupled-centered:** $a_{32} = 1$, $b_2 = 0$, and $\mathfrak{J}_\lambda = O(\lambda^{-4}) \rightarrow 0$.
3. **Case C, coupled-off-centered:** $a_{32} = 1$, $b_2 = 1$, and $\mathfrak{J}_\lambda \rightarrow b_2^2/\varepsilon_0 = 1$.

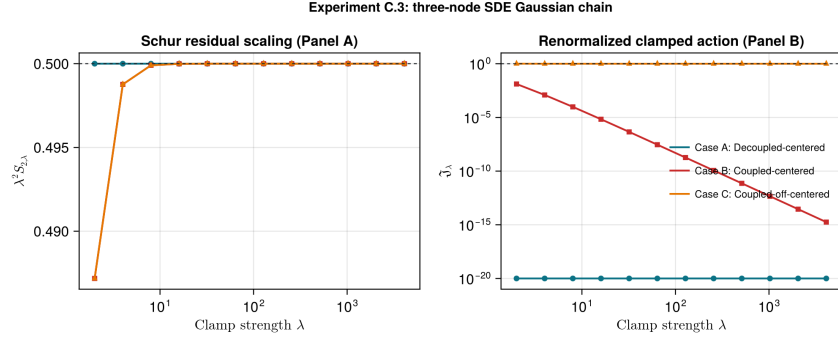


Figure 3: The panels show $\lambda^2 S_{2,\lambda} \rightarrow 0.5$ and the three finite-part regimes: exact zero, $O(\lambda^{-4})$ decay, and convergence to 1.

1.4 Cameron–Martin Admissibility Stress Test

For a 1D elliptic Gaussian field, smooth compactly supported sensors have finite Cameron–Martin energy. Approximating a point sensor by a concen-

trated mollifier g_ε drives the energy norm to infinity as $\varepsilon \downarrow 0$. This stress test illustrates the admissibility boundary in the paper’s theorem: smoothed observations are admissible, while the unsmoothed Dirac point-evaluation limit is not a finite-energy channel for the shorting theorem.

1.5 Sparse Precision-Operator Implementation

The diagnostics also include the OU framework with drift $\mathcal{A}_0 = -L^{-1}$ and formal covariance $Q_0 = L$ on the dual space $H^{-1}(0, 1)$. The resolvent admits the sparse conjugate realization

$$(\lambda I + L^{-1})^{-1} = L(\lambda L + I)^{-1} = \frac{1}{\tilde{\lambda}} L(L + \tilde{\lambda} I)^{-1}, \quad \tilde{\lambda} = \lambda^{-1}.$$

Thus the resolvent is obtained by a single sparse tridiagonal solve with the precision operator L at the conjugate parameter $\tilde{\lambda}$, avoiding dense covariance construction.

Table 3: Sparse precision-operator validation in the OU regime, $N = 10000$, $\kappa = 1$, $\Delta_E = 1$.

λ	$\tilde{\lambda} = \lambda^{-1}$	$\lambda^4 \text{tr}(\rho_{\lambda,E})$
10^1	10^{-1}	1.0263×10^{-7}
10^2	10^{-2}	1.0283×10^{-7}
10^3	10^{-3}	1.0285×10^{-7}
10^4	10^{-4}	1.0285×10^{-7}
10^5	10^{-5}	1.0285×10^{-7}

1.6 Precision Audit

At $n = 256$, representative Schur residuals were recomputed using IEEE 754 `Float64` and 256-bit Julia `BigFloat` arithmetic with the same covariance equation and Schur assembly.

For the five-point extrapolation of $m_2^{\text{Schur},(256)}$, `Float64` and `BigFloat` differ by 9.98×10^{-17} ; the finite- λ remainder, not rounding, governs the observed distance to the limiting constant.

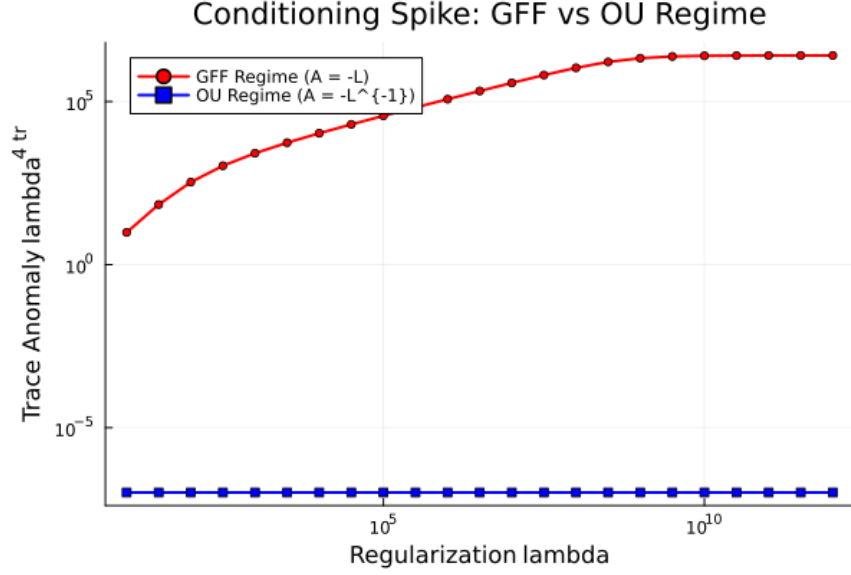


Figure 4: Trace anomaly conditioning comparison between the GFF and OU regimes.

Table 4: Precision audit at $n = 256$; S_{256} uses 256-bit **BigFloat**.

λ	$\kappa_2(\Sigma_{II})$	$ \lambda^2 S - \frac{1}{2} $	$ S_{64} - S_{256} / S_{256} $
4	2.97×10^6	3.45×10^{-1}	6.50×10^{-16}
64	9.14×10^5	1.13×10^{-3}	3.48×10^{-17}
4096	9.14×10^5	9.20×10^{-10}	5.54×10^{-17}